WebAssembly Distributed Dynamic Inference

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Thesis to obtain the Master of Science Degree in

Computer Science and Engineering

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Rodrigo Rodrigues

Set the examination month and year (e.g., February 2026).

Declaration
I declare that this document is an original work of my own authorship and that it fulfills all the requirements of the Code of Conduct and Good Practices of the Universidade de Lisboa.

Acknowledgments

I would like to thank my parents for their friendship, encouragement and caring over all these years, for always being there for me through thick and thin and without whom this project would not be possible. I would also like to thank my grandparents, aunts, uncles and cousins for their understanding and support throughout all these years.

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I would also like to acknowledge my dissertation supervisors Prof. Some Name and Prof. Some Other Name for their insight, support and sharing of knowledge that has made this Thesis possible.

Last but not least, to all my friends and colleagues that helped me grow as a person and were always there for me during the good and bad times in my life. Thank you.

To each and every one of you – Thank you.

Abstract

WebAssembly (Wasm) has emerged as a portable compilation target for multiple programming languages, extending beyond its original focus on web applications to domains like cloud computing. Despite its growing adoption, support for AI model inference in Wasm, particularly on serverless platforms like Cloudflare Workers [**?**], remains limited. The introduction of the WebAssembly Component Model provides a shared-nothing abstraction for modules, enabling interoperability between programming languages through memory encapsulation and standardized type specifications (WIT). Within this framework, interfaces like wasi-nn have been proposed to facilitate AI model inference using common backends such as PyTorch [**?**] and TensorFlow [**?**]. However, existing implementations of wasi-nn overlook cost optimization strategies, particularly those related to GPU rental.

The widespread adoption of deep learning models has driven significant technological advancements across domains such as healthcare, developer productivity, data analytics, and education. However, executing inference on infrequently accessed models, such as those hosted on platforms like Hugging Face, presents cost-efficiency challenges. Renting high-performance GPUs, like H100 or RTX 4090, for low-throughput tasks is economically impractical. Existing solutions like DynamoLLM [**?**] reduce costs by scaling GPU frequency and instances but lack support for heterogeneous vertical scaling.

This work addresses these limitations by proposing a runtime capable of dynamically scaling both vertically and horizontally based on cost-benefit analyses. Our approach improves throughput per dollar by renting cheaper resources, such as RTX 4090 GPUs or CPU VMs, for low-demand inference, while seamlessly scaling to meet production throughput requirements. Implementing this adaptive switching between CPU and GPU inference with existing frameworks like PyTorch requires overcoming challenges related to their distinct shared libraries and model weights management.

By incorporating cost-benefit analysis, our runtime will support vertical scaling—switching between GPU types, such as from an RTX 4090 for testing to an H100 for production—alongside horizontal scaling. This approach provides cost-efficient deployment options, from initial testing to production, and seamlessly integrates with existing Wasm applications via the standardized WebAssembly Component

Model.

Keywords

Maecenas tempus dictum libero; Donec non tortor in arcu mollis feugiat;Cras rutrum pulvinar tellus.

Resumo

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Palavras Chave

Colaborativo; Codificaçãoo; Conteúdo Multimédia; Comunicação;

Contents

List of Figures

List of Tables

List of Algorithms

Listagens

Acronyms

AVC	Advanced Video Coding
CC	Cloud Computing
CDN	Content Distribution Network
CPU	Central Processing Unit
DASH	Dynamic Adaptive Streaming over HTTP
GPRS	General Packet Radio Service
HD	High Definition
HTTP	Hypertext Transfer Protocol
LAN	Local Area Network
LTE	Long Term Evolution
OS	Operating System
SD	Standard Definition
SVC	Scalable Video Coding
UMTS	Universal Mobile Telecommunication System
WLAN	Wireless Local Area Network
WWAN	Wireless Wide Area Network

Glossary

formula

A mathematical expression 4

LaTeX

LaTeX It is a mark up language specially suited for scientific documents as it can correctly format documents with all the typographical rules 4

mathematics

Mathematics is what mathematicians do 4

1

Introduction

Contents

Rui Cruz: The examples of techniques, tools, and packages along the document are for you to get familiarized with them. It is advisable to preserve those examples of usage, for reference, by moving the respective blocks of text to the last Chapter of this template (or to a Chapter file that you know you will not use), until you finish your document.

Example of using package `todo` for notes of authors. In this case the author Johnny is calling the attention for something at the specific place in the text.

Johnny pointing out to the place

In this other case, another co-author is commenting on something inline.

Manuel: Inline comment or Note. It can be an extract of some recommended text. “Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Morbi commodo, ipsum sed pharetra gravida, orci magna rhoncus neque, id pulvinar odio lorem non turpis. Nullam sit amet enim. Suspendisse id velit vitae ligula volutpat condimentum. Aliquam erat volutpat. Sed quis velit. Nulla facilisi. Nulla libero. Vivamus pharetra posuere sapien.”

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Pete

You should cite also Pellentesque:2014

graphic record [1–3].

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Rui Cruz

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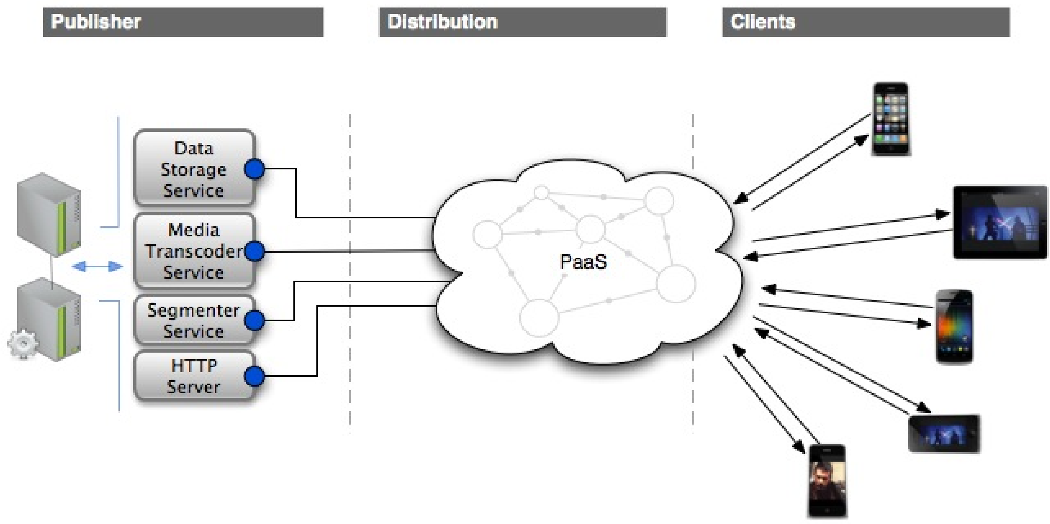

Figure 1.1: Ecosystem

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RC example of use of Glossaries

Notice that mathematics makes extensive use of Formulas which are particularly well rendered in documents produced with LaTeX.

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1.2 Organization of the Document

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2

This is the Second Chapter

Contents

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2.1 Traditional Streaming Technologies

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Sed pulvinar, “felis id consectetuer” malesuada, enim nisl mattis elit, a facilisis tortor nibh quis leo Table 2.1.

Table 2.1: Streaming Technologies Comparison

	Dynamic Streaming	Smooth Streaming	HLS
Streaming Protocol	RTMP	HTTP	HTTP
Video Codec	H.264, VP6	H.264	H.264
Audio Codec	AAC, MP3	WMA, AAC	AAC, MP3
Container Format	MP4, FLV,	MP4	MPEG2-TS
iOS	NO	YES	YES
Android	NO	YES	YES

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Table 2.2: A nice Spreadsheet using package “spreadtab”. Notice the calculations.

22	54	76
43	65	108
49	37	86
114	156	270

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Table 2.3: Comparison between today's and target Architectures of Telcos

Today		Target	
Rigid	Each evolutionary requirement involves development of multiple components, interfaces, platforms,etc.	Flexible	It is possible to modify or add new functionalities rapidly.
Slow	Development of a new application takes months or years.	Fast	Development of a new application takes weeks instead of months or years.
Closed	Limited integration with external environments.	Open	It is simple to integrate internal, applications with external entities.
Complex	Heterogeneous technologies, obsolescence, lack,of standards, high redundancy.	Standardised	Use of homogeneous architectural models.
Expensive	High Capex (for new service development) and,high,Opex (to ensure running of IT).	Cost-Effective	Capex and Opex are optimised.

3

This is the Third Chapter

Contents

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3.1 Architecture Design

Example of a Flowchart for a system, in Figure 3.1, created with `https://app.diagrams.net` and then exported as "PDF" crop format (a true vector image that can be scaled to no end, with no pixels or distortion).

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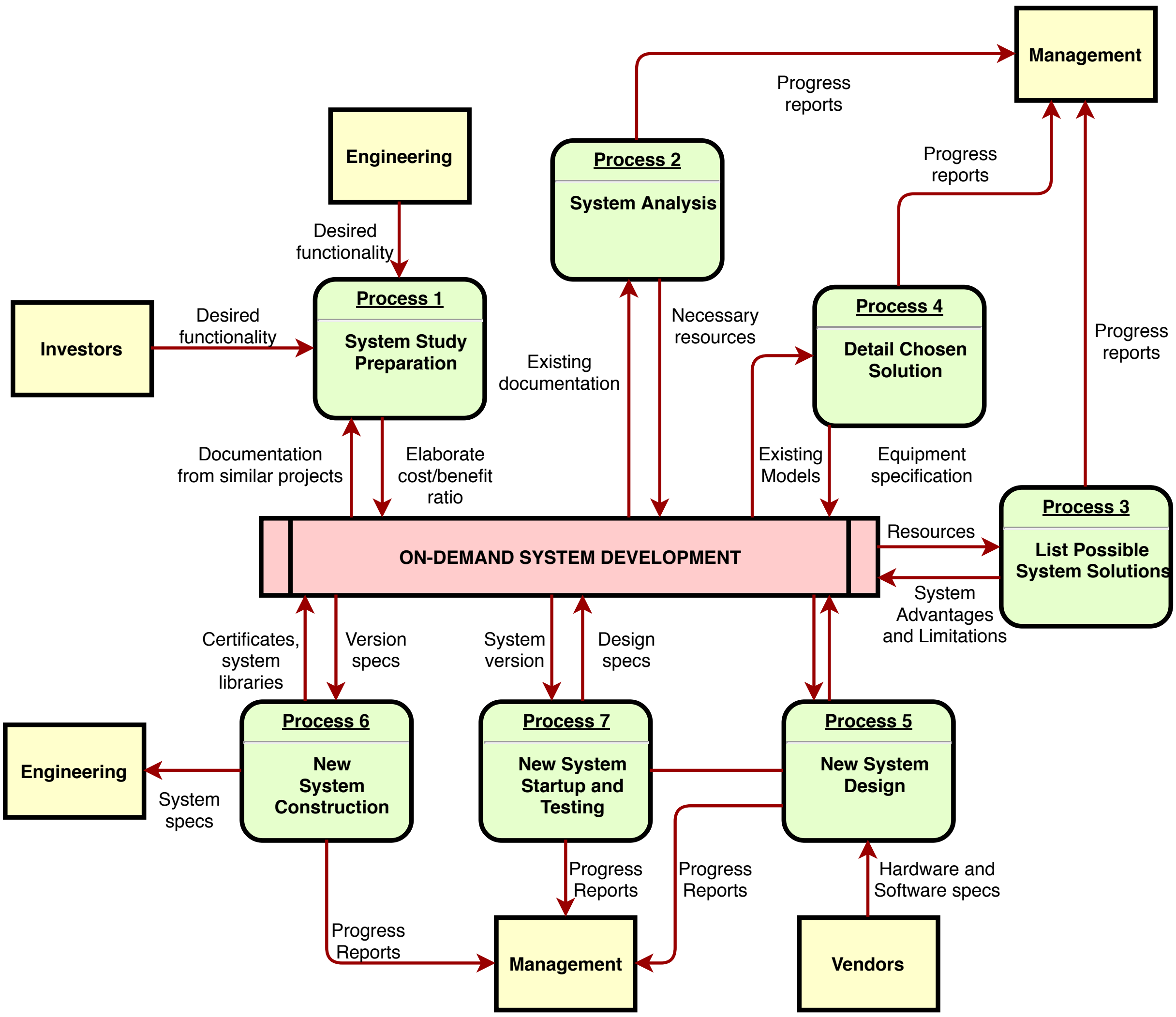

Figure 3.1: System Processes

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And here another diagram of a network (Figure 3.2) created with `https://app.diagrams.net` and then exported as "PDF" crop format.

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Web-streaming: The client application should support streaming media using Hypertext Transfer

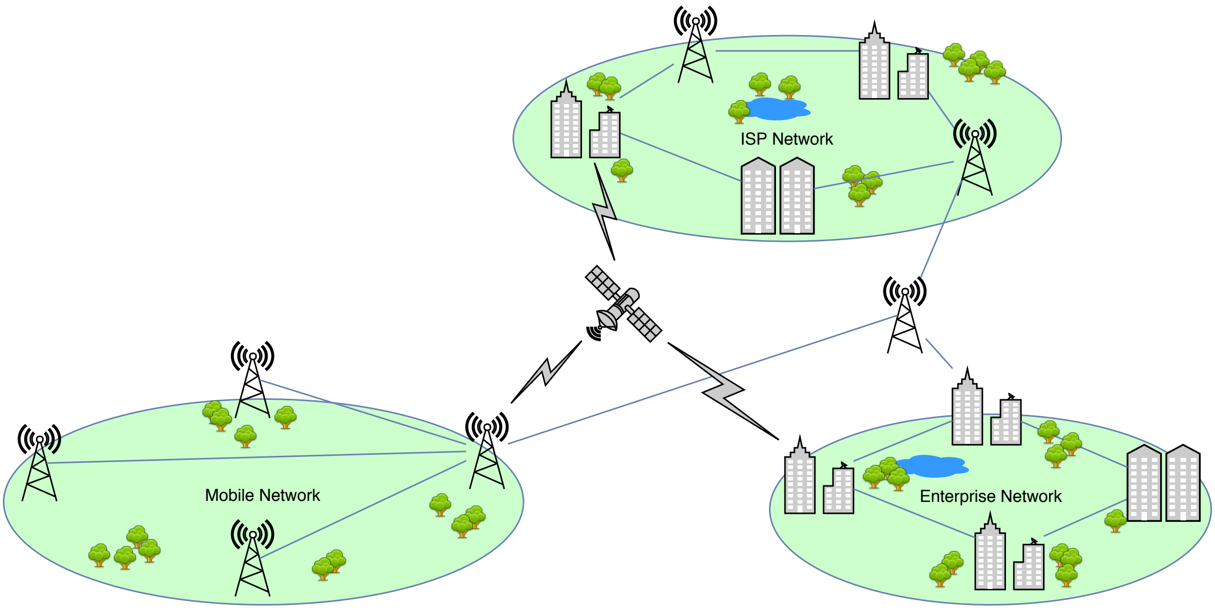

Figure 3.2: Network Diagram

Protocol (HTTP) protocols.

Multi-source streaming: The client application should support multi-source streaming media, i.e., "simultaneous" streaming of media content components from a network, supported/complemented by Content Distribution Network (CDN)/Cloud Computing (CC) services.

Support content Metadata Description: The client application should support content metadata description in a format similar or compliant with MPEG Dynamic Adaptive Streaming over HTTP (DASH) [11].

Scalable and Adaptive Media Contents: The system should support on-demand streaming of scalable and adaptive contents based on SVC.

Heterogenous End-User Devices: The client application should be compatible with current and future generations of end-user devices form factors, irrespective of their performance, screen size and resolution.

Access Network independency: The solution should provide the expected service over different types of access networks supported by the end-user devices, such as Wireless Local Area Networks (LANs) (IEEE 802.11) or cellular data networks such as General Packet Radio Service (GPRS), Universal Mobile Telecommunication System (UMTS), Long Term Evolution (LTE), etc.

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3.2 Architecture Design Requirements

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RC

A listing for XML code, with syntax highlighting

Listagem 3.1: Example of a MPD file.

```xml
<?xml version="1.0" encoding="UTF-8"?>
<StreamInfo version="2.0">
    <Clip duration="PT01M0.00S">
        <BaseURL>videos/</BaseURL>
        <Description>svc_1</Description>
        <Representation mimeType="video/SVC" codecs="svc" frameRate="30.00" bandwidth="401.90"
            width="176" height="144" id="L0">
            <BaseURL>svc_1/</BaseURL>
            <SegmentInfo from="0" to="11" duration="PT5.00S">
                <BaseURL>svc_1-L0-</BaseURL>
            </SegmentInfo>
        </Representation>
        <Representation mimeType="video/SVC" codecs="svc" frameRate="30.00"
             bandwidth="1322.60"
            width="352" height="288" id="L1">
            <BaseURL>svc_1/</BaseURL>
            <SegmentInfo from="0" to="11" duration="PT5.00S">
                <BaseURL>svc_1-L1-</BaseURL>
            </SegmentInfo>
        </Representation>
    </Clip>
</StreamInfo>
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4

Solution

Contents

Our solution to improve throughput per unitary cost when doing model inference first starts by being able to dynamically choose between different types of hardware with a variance of unitary cost per hour, for this we show an initial simplified version that only selects between a CPU VM and one type of rented GPU, but this should easily be expanded to a bigger variety of hardware accelerators. When doing inference however, to easily switch between the accelerators without shutting down the microservice running the code, we choose to run the main application code separate from the model inference on the hardware accelerator, and thus we require a way to link the our interface to a possibly local execution in the case of CPU inference, or remote execution for GPU inference.

4.0.1 Dynamic Module Selection

To optimize throughput per unitary cost, we select different hardware accelerators based on current throughput and utilization characteristics.

4.0.1.A High Throughput and High Device Utilization (HDU)

When the system operates at high throughput and thus high device utilization (HDU), GPUs should generally the optimal choice. GPUs should provide at high device utilization superior throughput per unitary cost and per watt (W) due to their parallel processing capabilities and the inherent parallelizability of deep learning model inference and training. This makes them always suitable for training models, where high device utilization is a common scenario.

4.0.1.B Low Throughput and Low Device Utilization (LDU)

In contrast, low throughput and low device utilization (LDU) may be encountered during inference, especially for rarely accessed models like those hosted on platforms such as Hugging Face. In these scenarios, CPUs emerge as the more cost-effective option. CPUs should offer better throughput per unitary cost in LDU conditions because they consume less power when idle or underutilized, and their energy requirements do not scale as drastically as GPUs.

Considering energy consumption from a holistic perspective—factoring in the energy used by the orchestrator or main CPU server in addition to the GPU—CPUs may also demonstrate higher throughput per watt. This is particularly true in architectures where separate servers are used for orchestration and computation.

To implement this we obtain throughput per unitary cost profiles of each of the possible hardware types from a historical throughput profiles data source and select the lowest unitary cost for our current inference throughput, we outline the final architecture for this dynamic selection in section 4.0.3.

As we discussed in section 4, our solution separates the inference hardware accelerators from the main microservice component to prevent it from being shutdown, thus we require a way to link these the main microservice component with the inference hardware accelerator components, to do this we propose an initial solution based on dynamic linking with WebAssembly primitives, and then improve on this initial proposal's performance bottlenecks by using modern WebAssembly component model infrastructure.

4.0.2 Remote Module Execution with Dynamic Linking

Our initial solution for remote module execution uses WASI v1 and Wasmtime runtime abstractions, such as **Store**, **Linker**, and **Module**, to implement dynamic linking. This involves manually handling shared libraries, memory, and tables.

4.0.2.A Application Compilation

To create these modules, we use the WASI SDK compiler toolchain with -fPIC, -shared and -Wl,–export-all flags to compile our shared libraries. The -fPIC Flag makes sure the compiler generates position independent WASM code, the -shared flag generates the shared library and the -Wl,–export-all flag makes sure the linker exports all symbols even if unused. After compiling the shared libraries, we use also compile the main module as a shared library.

4.0.2.B Dynamic Linking Approaches

There are multiple approaches to dynamic linking, such as shared-nothing dynamic linking, shared-something dynamic linking, and shared-everything dynamic linking.

4.0.2.C Shared-Nothing Dynamic Linking

Shared-nothing dynamic linking architecture is an approach to dynamic linking that does not allow for the sharing of resources between modules other than with explicit communication. This approach is used in the Component Model, which is a part of the WebAssembly System Interface (WASI) v2 standard.

4.0.2.D Dynamic Linking Implementation

We base our dynamic linking implementation on existing work [**?**] that was done for the EMScripten compiler toolchain and not the WASI SDK compiler toolchain.

4.0.2.E Remote Module Execution

This solution for remote module execution, as can be seen in figure 4.1, brings significant performance bottlenecks due to shared memory between the servers.

4.0.3 Remote Module Execution with Component Model and wRPC

The second solution represented in Figure 4.2, builds on the Component Model and wRPC, by enforcing **shared-nothing** principles and leveraging the immutable high-level WIT interfaces from the component model, this approach eliminates the need of sharing memory between the components.

- Components communicate explicitly via immutable WIT interfaces, and each component has it's own linear memory.
- Remote execution is enabled through wRPC

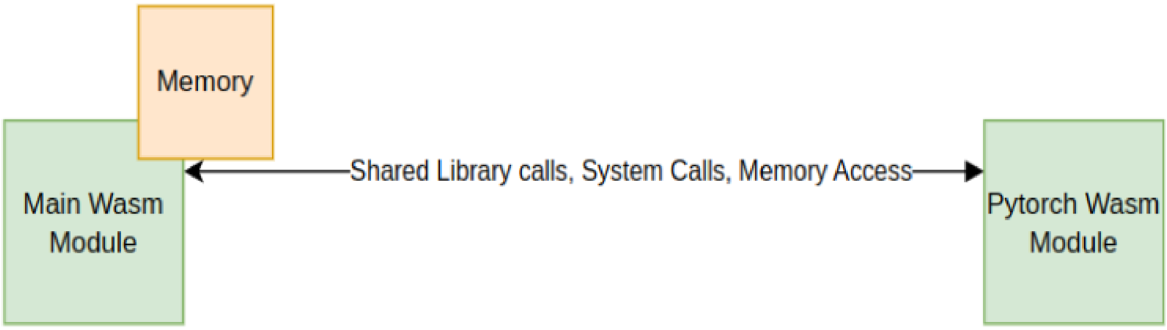

Figure 4.1: Architecture for remote module execution with shared memory.

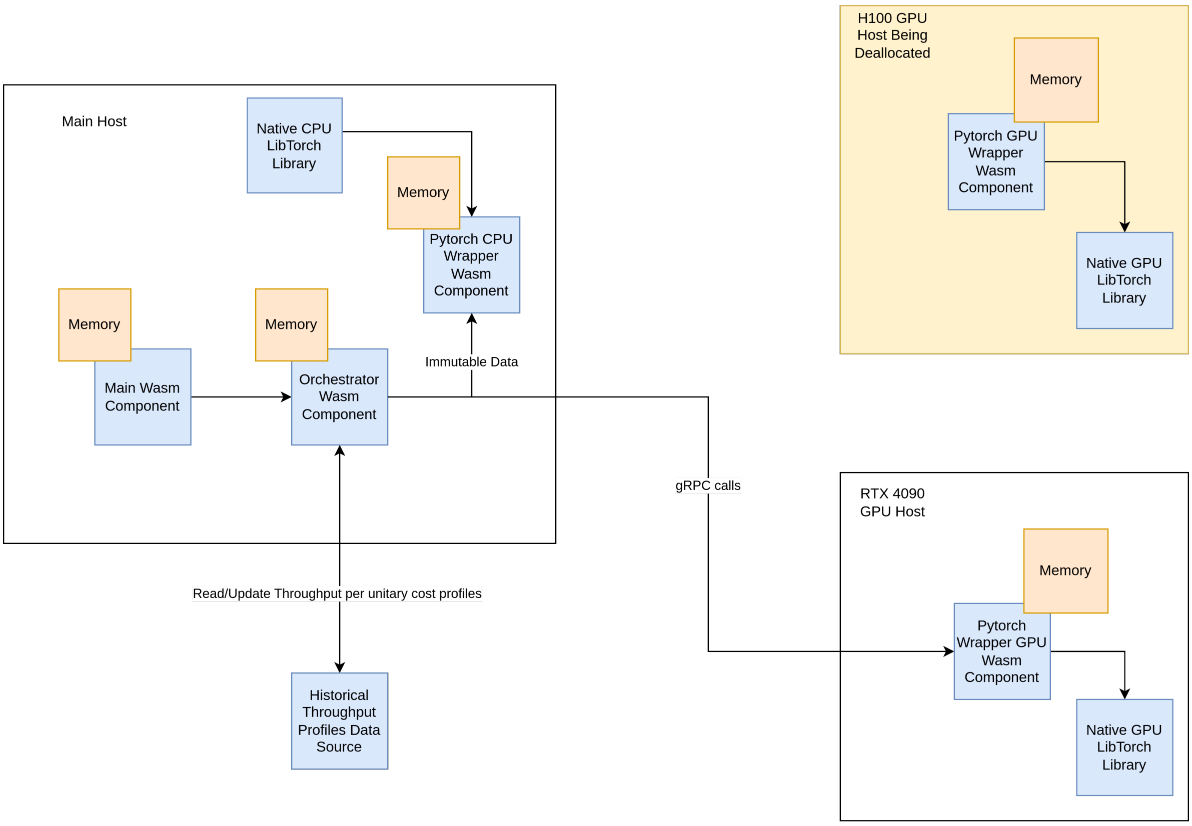

Figure 4.2: System architecture for dynamic selection of accelerators based on workload.

By dynamically selecting the appropriate accelerator based on throughput and device utilization, we believe lower costs and possible energy savings can be achieved.

5

This is the Fifth Chapter

Contents

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Morbi commodo, ipsum sed pharetra gravida, orci magna rhoncus neque, id pulvinar odio lorem non turpis. Nullam sit amet enim. Suspendisse id velit vitae ligula volutpat condimentum. Aliquam erat volutpat. Sed quis velit. Nulla facilisi. Nulla libero. Vivamus pharetra posuere sapien. Nam consectetuer. Sed aliquam, nunc eget euismod ullamcorper, lectus nunc ullamcorper orci, fermentum bibendum enim nibh eget ipsum. Donec porttitor ligula eu dolor. Maecenas vitae nulla consequat libero cursus venenatis. Nam magna enim, accumsan eu, blandit sed, blandit a, eros.

5.1 Maecenas vitae nulla consequat

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Sed sollicitudin velit eu magna. Aliquam erat volutpat. Vivamus ornare est non wisi. Proin vel quam. Vivamus egestas. Nunc tempor diam vehicula mauris. Nullam sapien eros Figure 5.1, facilisis vel, eleifend non, auctor dapibus, pede.

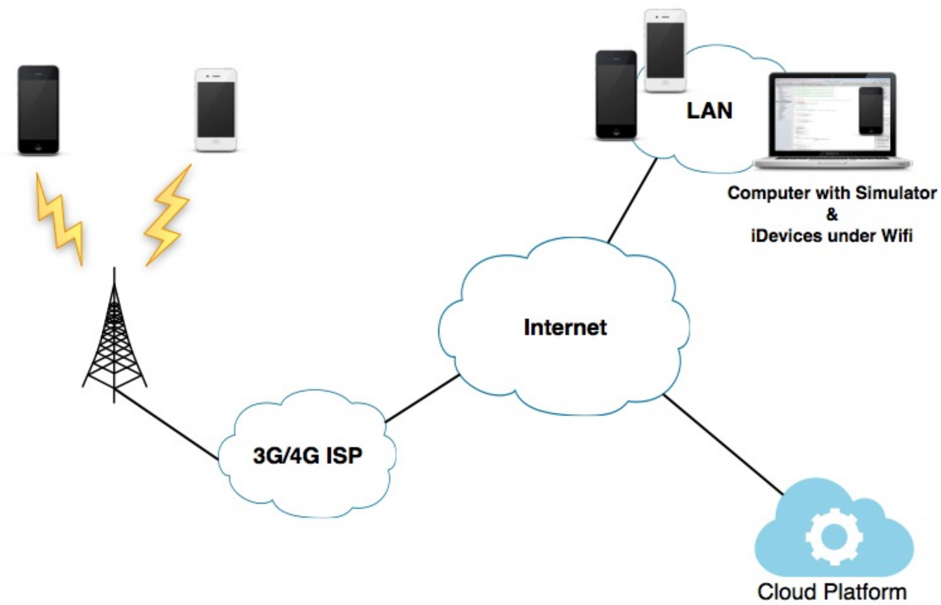

Figure 5.1: Test Environment

Aliquam aliquet, est a ullamcorper condimentum, tellus nulla fringilla elit, a iaculis nulla turpis sed wisi. Fusce volutpat. Etiam sodales ante id nunc. Proin ornare dignissim lacus. Nunc porttitor nunc a sem. Sed sollicitudin velit eu magna. Aliquam erat volutpat. Vivamus egestas. Nunc tempor diam vehicula mauris. Nullam sapien eros, facilisis vel, eleifend non, auctor dapibus, pede Table 5.1 used in the tests. The Network Link Conditioner allows to force/simulate fluctuations in fixed network segments.

Table 5.1: Network Link Conditioner Profiles

Network Profile	Bandwidth	Packets Droped	Delay
Wifi	40 mbps	0%	1 ms
3G	780 kbps	0%	100 ms
Edge	240 kbps	0%	400 ms

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5.2 Proin ornare dignissim lacus

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Et "optimistic" nulla dui purus, eleifend vel, consequat non, dictum porta, nulla. Duis ante mi, laoreet ut, commodo eleifend, cursus nec, lorem. Aenean eu est. Etiam imperdiet turpis. Praesent nec augue. Curabitur ligula quam, rutrum id, tempor sed, consequat ac, dui G_j, nec ligula et lorem consequat ullamcorper p ut mauris eu mi mollis luctus j, porttitor ut, Equation (5.1), uctus posuere justo:

N_j Is the number of times peer j has been optimistically unchoked.

n_j Among the N_j unchokes, the number of times that peer j responded with unchoke or supplied segments to peer p.

$C_{r[j]}$ The cooperation ratio of peer j. If peer j never supplied peer p, the information of $C_{r[j]}$ may not be available.

$C_{r(max)}$ The maximum cooperation ratio of peer p's neighbors, i.e., $C_{r(max)} = max(C_r)$.

$$G_j = \begin{cases} \dfrac{n_j C_{r[j]}}{N_j} & \text{if } n_j > 0 \\ \dfrac{C_{r(max)}}{N_j + 1} & \text{if } n_j = 0 \end{cases} \tag{5.1}$$

Cursus $C_{r(max)}$ conubia nostra, per inceptos hymenaeos j gadipiscing mollis massa $N_j = 0$, unc ut dui eget nulla venenatis aliquet $G_j = C_{r(max)}$.

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Both Figures 5.2(a) and 5.2(b) Phasellus eget nisl ut elit porta "perfect" tincidunt. Class aptent taciti sociosqu ad litora torquent per conubia nostra.

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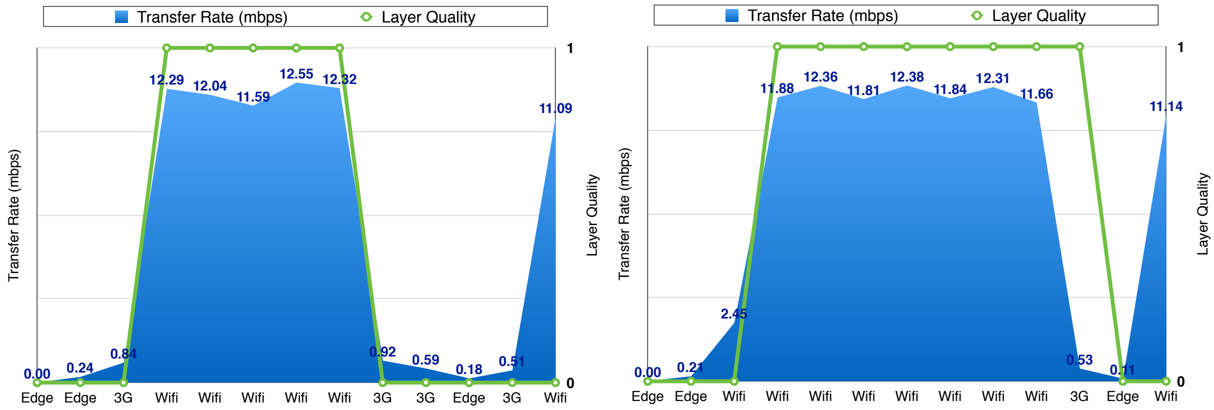

(a) Adaptation System Test 4

(b) Adaptation System Test 5

Figure 5.2: Adaptation System Behavior Test

Sed luctus posuere justo. Cras vehicula varius turpis. Vivamus eros metus, tristique sit amet, molestie dignissim, malesuada et, urna.

6

Conclusion

Contents

Pellentesque vel dui sed orci faucibus iaculis. Suspendisse dictum magna id purus tincidunt rutrum. Nulla congue. Vivamus sit amet lorem posuere dui vulputate ornare. Phasellus mattis sollicitudin ligula. Duis dignissim felis et urna. Integer adipiscing congue metus.

Rui Cruz
You should always start a Chapter with an introductory text

6.1 Conclusions

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6.2 System Limitations and Future Work

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A

Code of Project

Nulla dui purus, eleifend vel, consequat non, dictum porta, nulla. Duis ante mi, laoreet ut, commodo eleifend, cursus nec, lorem. Aenean eu est. Etiam imperdiet turpis. Praesent nec augue. Curabitur ligula quam, rutrum id, tempor sed, consequat ac, dui. Vestibulum accumsan eros nec magna. Vestibulum vitae dui. Vestibulum nec ligula et lorem consequat ullamcorper.

Listagem A.1: Example of a XML file.

```xml
<?xml version="1.0" encoding="UTF-8"?>
<StreamInfo version="2.0">
    <Clip duration="PT01M0.00S">
        <BaseURL>videos/</BaseURL>
        <Description>svc_1</Description>
        <Representation mimeType="video/SVC" codecs="svc" frameRate="30.00" bandwidth="401.90"
            width="176" height="144" id="L0">
            <BaseURL>svc_1/</BaseURL>
            <SegmentInfo from="0" to="11" duration="PT5.00S">
                <BaseURL>svc_1-L0-</BaseURL>
            </SegmentInfo>
        </Representation>
```

```xml
            <Representation mimeType="video/SVC" codecs="svc" frameRate="30.00" bandwidth="1322.60"
                width="352" height="288" id="L1">
                <BaseURL>svc_1/</BaseURL>
                <SegmentInfo from="0" to="11" duration="PT5.00S">
                    <BaseURL>svc_1-L1-</BaseURL>
                </SegmentInfo>
            </Representation>
        </Clip>
</StreamInfo>
```

Etiam imperdiet turpis. Praesent nec augue. Curabitur ligula quam, rutrum id, tempor sed, consequat ac, dui. Maecenas tincidunt velit quis orci. Sed in dui. Nullam ut mauris eu mi mollis luctus. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Sed cursus cursus velit. Sed a massa. Duis dignissim euismod quam.

Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Phasellus eget nisl ut elit porta ullamcorper. Maecenas tincidunt velit quis orci. Sed in dui. Nullam ut mauris eu mi mollis luctus. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos.

This inline MATLAB code `for i=1:3, disp('cool'); end;` uses the `\mcode{}` command.[1]

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Listagem A.2: `Matlab Function`

```matlab
for i = 1:3
    if i >= 5 && a ~= b        % literate programming replacement
        disp('cool');          % comment with some LaTeXin it:  πx^2
    end
    [:,ind] = max(vec);
    x_last = x(1,end) - 1;
    v(end);
    ylabel('Voltage (µV)');
end
```

Nullam ut mauris eu mi mollis luctus. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Sed cursus cursus velit. Sed a massa. Duis dignissim euismod quam. Nullam euismod metus ut orci.

[1] MATLAB Works also in footnotes: `for i=1:3, disp('cool'); end;`

Listagem A.3: `function.m`

```
%   Copyright 2010 The MathWorks, Inc.
function ObjTrack(position)
% #codegen
% First, setup the figure
numPts = 300;                % Process and plot 300 samples
figure;hold;grid;            % Prepare plot window
% Main loop
for idx = 1: numPts
    z = position(:,idx);     % Get the input data
    y = kalmanfilter(z);     % Call Kalman filter to estimate the position
    plot_trajectory(z,y);    % Plot the results
end
hold;
end   % of the function
```

Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Phasellus eget nisl ut elit porta ullamcorper. Maecenas tincidunt velit quis orci. Sed in dui. Nullam ut mauris eu mi mollis luctus. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Sed cursus cursus velit. Sed a massa. Duis dignissim euismod quam. Nullam euismod metus ut orci. Vestibulum erat libero, scelerisque et, porttitor et, varius a, leo.

Listagem A.4: HTML with CSS Code

```
<!DOCTYPE html>
<html>
  <head>
    <title>Listings Style Test</title>
    <meta charset="UTF-8">
    <style>
      /* CSS Test */
      * {
        padding: 0;
        border: 0;
        margin: 0;
      }
    </style>
    <link rel="stylesheet" href="css/style.css" />
  </head>
```

```
<header> hey </header>
<article> this is a article </article>
<body>
  <!-- Paragraphs are fine -->
  <div id="box">
    <p>
      Hello World
    </p>
    <p>Hello World</p>
    <p id="test">Hello World</p>
    <p></p>
  </div>
  <div>Test</div>
  <!-- HTML script is not consistent -->
  <script src="js/benchmark.js"></script>
  <script>
    function createSquare(x, y) {
      // This is a comment.
      var square = document.createElement('div');
      square.style.width = square.style.height = '50px';
      square.style.backgroundColor = 'blue';

      /*
       * This is another comment.
       */
      square.style.position = 'absolute';
      square.style.left = x + 'px';
      square.style.top = y + 'px';

      var body = document.getElementsByTagName('body')[0];
      body.appendChild(square);
    };

    // Please take a look at +=
    window.addEventListener('mousedown', function(event) {
      // German umlaut test: Berührungspunkt ermitteln
      var x = event.touches[0].pageX;
      var y = event.touches[0].pageY;
```

```
            var lookAtThis += 1;
          });
        </script>
      </body>
    </html>
```

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Listagem A.5: HTML CSS Javascript Code

```

@media only screen and (min-width: 768px) and (max-width: 991px) {

  #main {
    width: 712px;
    padding: 100px 28px 120px;
  }

  /* .mono {
    font-size: 90%;
  } */

  .cssbtn a {
    margin-top: 10px;
    margin-bottom: 10px;
    width: 60px;
    height: 60px;
    font-size: 28px;
    line-height: 62px;
  }
```

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Listagem A.6: PYTHON Code

```
class TelgramRequestHandler(object):
    def handle(self):
        addr = self.client_address[0]          # Client IP-adress
        telgram = self.request.recv(1024)      # Recieve telgram
        print "From: %s, Received: %s" % (addr, telgram)
        return
```

B

A Large Table

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As Table B.1 shows, the data can be inserted from a file, in the case of a somehow complex structure. Notice the Table footnotes.

Table B.1: Example table

Benchmark: ANN	**#Layers** (1)	**#Nets** (2)	#Nodes* $(3) = 8 \cdot (1) \cdot (2)$	Critical path $(4) = 4 \cdot (1)$	**Latency (T_{iter})** (5)
A1	**3–1501**	1	**24–12008**	**12–6004**	4
A2	501	1	4008	2004	**2–2000**
A3	10	**2–1024**	**160–81920**	40	60†
A4	10	50	4000	40	**80–1200**

Benchmark: FFT	**FFT size‡** (1)	*#Inputs* $(2) = 2^{(1)}$	*#Nodes** $(3) = 10 \cdot (1) \cdot (2)$	*Critical path* $(4) = 4 \cdot (1)$	**Latency (T_{iter})** (5)
F1	**1–10**	2–1024	**20–102400**	4–40	6–60†
F2	**5**	32	1600	20	**40 – 1500**

Benchmark: Random networks	**#Types** (1)	**#Nodes** (2)	*#Networks* (3)	*Critical path* (4)	**Latency (T_{iter})** (5)
R1	3	10–2000	500	*variable*	(4)
R2	3	50	500	*variable*	$(4) \times [1; \cdots ; 20]$

* *Excluding constant nodes.*
† *Value kept proportional to the critical path:* $(5) = (4) * 1.5$.
‡ *A size of* x *corresponds to a* 2^x *point FFT.*
Values in bold indicate the parameter being varied.

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And now an example (Table B.2) of a table that extends to more than one page. Notice the repetition of the Caption (with indication that is continued) and of the Header, as well as the continuation text at the bottom.

Table B.2: Example of a very long table spreading in several pages

Time (s)	**Triple chosen**	**Other feasible triples**
0	(1, 11, 13725)	(1, 12, 10980), (1, 13, 8235), (2, 2, 0), (3, 1, 0)
2745	(1, 12, 10980)	(1, 13, 8235), (2, 2, 0), (2, 3, 0), (3, 1, 0)
5490	(1, 12, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
8235	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
10980	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
13725	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
16470	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)

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Table B.2 – continued from previous page

Time (s)	Triple chosen	Other feasible triples
19215	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
21960	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
24705	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
27450	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
30195	(2, 2, 2745)	(2, 3, 0), (3, 1, 0)
32940	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
35685	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
38430	(1, 13, 10980)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
41175	(1, 12, 13725)	(1, 13, 10980), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
43920	(1, 13, 10980)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
46665	(2, 2, 2745)	(2, 3, 0), (3, 1, 0)
49410	(2, 2, 2745)	(2, 3, 0), (3, 1, 0)
52155	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
54900	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
57645	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
60390	(1, 12, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
63135	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
65880	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
68625	(2, 2, 2745)	(2, 3, 0), (3, 1, 0)
71370	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
74115	(1, 12, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
76860	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
79605	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
82350	(1, 12, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
85095	(1, 12, 13725)	(1, 13, 10980), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
87840	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
90585	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
93330	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
96075	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
98820	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
101565	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
104310	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
107055	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
109800	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
112545	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
115290	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
118035	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
120780	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
123525	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
126270	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
129015	(2, 2, 2745)	(2, 3, 0), (3, 1, 0)
131760	(2, 2, 2745)	(2, 3, 0), (3, 1, 0)
134505	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
137250	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
139995	(2, 2, 2745)	(2, 3, 0), (3, 1, 0)
142740	(2, 2, 2745)	(2, 3, 0), (3, 1, 0)
145485	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
148230	(2, 2, 2745)	(2, 3, 0), (3, 1, 0)
150975	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
153720	(1, 12, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)

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Table B.2 – continued from previous page

Time (s)	Triple chosen	Other feasible triples
156465	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
159210	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
161955	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
164700	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)

An example of a large Table that autofits the size to the page margins is illustrated in Table B.3. Please notice the text size that is shrunken in order fot the table to adjust to the page:

Table B.3: Sample Table.

URL	First Time Visit	Last Time Visit	URL Counts	Value	Reference
https://web.facebook.com/	1521241972	1522351859	177	56640	[facebook-2021]
http://localhost/phpmyadmin/	1518413861	1522075694	24	39312	database-management
https://mail.google.com/mail/u/	1516596003	1522352010	36	33264	Google-Gmail-2021
https://github.com/shawon100	1517215489	1522352266	37	27528	Code-Repository
https://www.youtube.com/	1517229227	1521978502	24	14792	Youtube-video-2021